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AEROSOL GENERATING DEVICE AND HEATING STRUCTURE THEREOF

Abstract

A heating structure includes: a heating element for radiating an infrared wave in a powered-on state; and a tube body disposed in conjunction with the heating element, the tube body allowing the infrared wave to pass through. The heating element is at least partially spaced apart from a tube wall of the tube body. The heating element includes a heating part and a conductive part. The heating part includes at least two heating sections arranged spaced apart. The heating sections are electrically connected to the conductive part.

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Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATION [0001] This application is a continuation of International Patent Application No. PCT/CN2023/114112, filed on Aug. 21, 2023, which claims priority to Chinese Patent Application No. 202223072237.X, filed on Nov. 17, 2022. The entire disclosure of both applications is hereby incorporated by reference herein.

FIELD

[0002] The present disclosure relates to the field of heat-not-burn atomization, and more specifically, to an aerosol generating device and a heating structure thereof.

BACKGROUND

[0003] In the related art, an aerosol generating device is an electronic device configured to heat an aerosol generating substrate (a solid substrate such as a plant leaf product, like tobacco) but not to combust the aerosol generating substrate. Generally, in the aerosol generating device, heating is performed at the center or the periphery of a heating element. Generally, the heating element is powered up to generate heat, and then the heat is directly transferred to an aerosol generating substrate through the heat conduction. The aerosol generating substrate is generally atomized within 350° C. The disadvantage of this heating manner is that the heating element directly transfers or indirectly transfers, by using a solid material through the heat conduction, the heat to the aerosol generating substrate. This requires that the working temperature of the heating element cannot be excessively high; otherwise, overheating of the substrate is to be caused, and the vaping taste of the e-cigarette is affected.

[0004] In addition, the low temperature of the heating element directly causes the long preheat time to be required for the aerosol generating device before the vaping is performed. Currently, the preheat time of a market product is basically 15 seconds or more, greatly affecting experience of a consumer. In addition, when the heating element is at the working temperature above 400° C., the aerosol generating substrate easily overheats, affecting the vaping taste. Therefore, how to properly configure a heating structure in a high-temperature working state is an urgent problem to be resolved in the art.

SUMMARY

[0005] In an embodiment, the present invention provides a heating structure, comprising: a heating element configured to radiate an infrared wave in a powered-on state; and a tube body disposed in conjunction with the heating element, the tube body being configured to allow the infrared wave to pass through, wherein the heating element is at least partially spaced apart from a tube wall of the tube body, wherein the heating element comprises a heating part and a conductive part, wherein the heating part comprises at least two heating sections arranged spaced apart, and wherein the heating sections are electrically connected to the conductive part.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached

drawings, which illustrate the following:

[0007] FIG. 1 is a schematic diagram of a three-dimensional structure of an aerosol generating device according to some embodiments of the present disclosure;

[0008] FIG. 2 is a schematic sectional view of a heating structure of the aerosol generating device shown in FIG. 1;

[0009] FIG. 3 is a schematic diagram of an exploded structure of the heating structure shown in FIG. 2;

[0010] FIG. 4 is a schematic diagram of a three-dimensional structure in which the heating element shown in FIG. 2 is wound, in an M shape, on a first framework, a second framework, a third framework, and a fourth framework that have different radial dimensions;

[0011] FIG. 5 is a schematic sectional view of a heating structure according to another embodiment of the present disclosure; and

[0012] FIG. 6 is a schematic diagram of an exploded structure of the heating structure shown in FIG. 5.

DETAILED DESCRIPTION

[0013] In an embodiment, the present invention provides an improved aerosol generating device and a heating structure thereof.

[0014] In an embodiment, the present invention provides a heating structure that includes a heating element capable of radiating an infrared wave in a powered-on state and a tube body that is disposed in conjunction with the heating element and that allows the infrared wave to pass through. The heating element is at least partially spaced apart from the tube wall of the tube body. The heating element includes a heating part and a conductive part. The heating part includes at least two heating sections arranged spaced apart. The heating sections are electrically connected to the conductive part.

[0015] In some embodiments, the heating structure further includes a framework, the framework is disposed on the tube body, and at least one of the heating sections is disposed on the framework.

[0016] In some embodiments, the framework includes a first framework, a second framework, a third framework, and a fourth framework. The at least two heating sections include a first heating section and a second heating section connected to the first heating section. The first framework and the second framework are spaced apart and separately disposed on the first heating section. The third framework and the fourth framework are spaced apart and separately disposed on the second heating section.

[0017] In some embodiments, the heating part is longitudinally disposed. The at least two heating sections include a first heating section and a second heating section connected to the first heating section. The first heating section and/or the second heating section are/is wound on the framework in an M shape or an N shape.

[0018] In some embodiments, the length of the first heating section is less than, equal to, or greater than the length of the second heating section.

[0019] In some embodiments, the first heating section and the second heating section include a plurality of M-shaped or N-shaped sections disposed at intervals.

[0020] In some embodiments, the plurality of M-shaped or N-shaped sections are formed by winding or bending strip-shaped or linear heating wires. The distribution density of the plurality of M-shaped or N-shaped sections of the first heating section on the framework is lower than, equal to, or higher than the distribution density of the plurality of M-shaped or N-shaped sections of the second heating section on the framework.

[0021] In some embodiments, the radial dimension of the first framework is equal to the radial dimension of the second framework, and the radial dimension of the third framework is smaller than, equal to, or larger than the radial dimension of the second framework.

[0022] In some embodiments, the radial dimension of the first framework is smaller than the radial dimension of the second framework, and the radial dimension of the third framework is smaller

than, equal to, or larger than the radial dimension of the second framework.

[0023] In some embodiments, the radial dimension of the first framework is larger than the radial dimension of the second framework, and the radial dimension of the third framework is smaller than, equal to, or larger than the radial dimension of the second framework.

[0024] In some embodiments, the first framework includes a plurality of first wire slots provided on the outer periphery edge of the first framework. The second framework includes a plurality of second wire slots corresponding to the plurality of first wire slots. The heating part is wound on the plurality of first wire slots and the plurality of second wire slots in an M shape or an N shape. The third framework includes a plurality of third wire slots provided on the outer periphery edge of the third framework. The fourth framework includes a plurality of fourth wire slots corresponding to the plurality of third wire slots. The heating part is wound on the plurality of third wire slots and the plurality of fourth wire slots in an M shape or an N shape.

[0025] In some embodiments, when the spacing between the plurality of first wire slots and the tube body is equal to the spacing between the plurality of second wire slots and the tube body, the spacing between the plurality of third wire slots and the tube body is greater than, less than, or equal to the spacing between the plurality of fourth wire slots and the tube body.

[0026] In some embodiments, when the spacing between the plurality of first wire slots and the tube body is less than the spacing between the plurality of second wire slots and the tube body, the spacing between the plurality of third wire slots and the tube body is greater than, less than, or equal to the spacing between the plurality of fourth wire slots and the tube body.

[0027] In some embodiments, when the spacing between the plurality of first wire slots and the tube body is greater than the spacing between the plurality of second wire slots and the tube body, the spacing between the plurality of third wire slots and the tube body is greater than, less than, or equal to the spacing between the plurality of fourth wire slots and the tube body.

[0028] In some embodiments, the heating part includes a heating substrate that generates heat in a powered-on state, and an infrared radiation layer that is disposed on the outer surface of the heating substrate and that is used for radiating the infrared wave.

[0029] In some embodiments, the heating structure further includes a base, the tube body is mounted on the base, and the conductive part of the heating element passes through the base.

[0030] In some embodiments, the heating element is disposed on the inner side of the tube body, and a spacing is defined between the heating element and the inner wall of the tube body.

[0031] In some embodiments, the tube body includes a first sleeve tube and a second sleeve tube sleeved on the outer periphery of the first sleeve tube.

[0032] The spacing is defined between the first sleeve tube and the second sleeve tube, and an accommodating cavity for accommodating the heating element is defined by the spacing.

[0033] The heating element is disposed on and spaced apart from the outer periphery of the first sleeve tube, and a heating cavity for heating an aerosol generating substrate is defined on the inner side of the first sleeve tube.

[0034] In some embodiments, a reflective layer reflecting the infrared wave is disposed on the inner wall of the second sleeve tube.

[0035] In some embodiments, when the spacing between the plurality of first wire slots and the first sleeve tube is equal to the spacing between the plurality of second wire slots and the first sleeve tube, the spacing between the plurality of third wire slots and the first sleeve tube is greater than, less than, or equal to the spacing between the plurality of fourth wire slots and the first sleeve tube.

[0036] In some embodiments, when the spacing between the plurality of first wire slots and the first sleeve tube is less than the spacing between the plurality of second wire slots and the first sleeve tube, the spacing between the plurality of third wire slots and the first sleeve tube is greater than, less than, or equal to the spacing between the plurality of fourth wire slots and the first sleeve tube.

[0037] In some embodiments, when the spacing between the plurality of first wire slots and the first sleeve tube is greater than the spacing between the plurality of second wire slots and the first sleeve tube, the spacing between the plurality of third wire slots and the first sleeve tube is greater than, less than, or equal to the spacing between the plurality of fourth wire slots and the first sleeve tube.

[0038] The present disclosure also provides an aerosol generating device, including the heating structure described in any one of the foregoing embodiments.

[0039] The present disclosure has the following beneficial effects: The heating element of the heating structure of the present disclosure can radiate an infrared wave in a powered-on state. The infrared wave may pass through the tube body to and heat the aerosol generating substrate. When the maximum working temperature of the heating element reaches 1000° C. or above (where the working temperature of the heating element of a conventional HNB generally does not exceed 400° C.), the overheating of the aerosol generating substrate is not caused, and the vaping taste can even be greatly improved. In addition, the preheat time is greatly shortened, thereby greatly improving experience of a consumer.

[0040] The heating part includes the at least two heating sections arranged spaced apart. Therefore, the heating sections can match different aerosol generating substrates and perform segmented heating, to provide proper temperature field distribution, so that the burning of a part of the substrate is avoided, and the vaping taste is further improved.

[0041] To bring a clearer understanding of the technical features, objectives, and effects of the present disclosure, specific implementations of the present disclosure are described with reference to the accompanying drawings.

[0042] In the descriptions of the present disclosure, it should be understood that orientation or position relationships indicated by terms such as “axial”, “length”, “upper”, “lower”, “top”, “inner”, and “outer” are orientation or position relationships shown based on the accompanying drawings or orientation or position relationships that the product of the present disclosure is usually placed in use, and are merely used for describing the present disclosure and simplifying the description, rather than indicating or implying that the mentioned apparatus or element should have a particular orientation or be constructed and operated in a particular orientation. Therefore, there terms should not be construed as a limitation to the present disclosure.

[0043] In addition, terms “first” and “second” are used merely for the purpose of description, and cannot be construed as indicating or implying relative importance or implying a quantity of indicated technical features. Therefore, features restricted by “first” or “second” may explicitly indicate or implicitly include at least one of such features. In the descriptions of the present disclosure, unless explicitly specified, “a plurality of” means at least two, for example, two or three.

[0044] In the present disclosure, it should be noted that unless otherwise explicitly specified and limited, the terms “mount”, “connect”, “connection”, and “fasten” should be understood in a broad sense. For example, a connection may be a fixed connection, a detachable connection, or an integral connection; or the connection may be a mechanical connection or an electrical connection; or the connection may be a direct connection, an indirect connection through an intermediary, or internal communication between two elements or mutual action relationship between two elements, unless otherwise specified explicitly. A person of ordinary skill in the art may understand specific meanings of the foregoing terms in the present disclosure according to specific situations.

[0045] FIG. 1 shows an aerosol generating device 1 and aerosol-generating substrate 2 detachably inserted into one end of the aerosol generating device 1 in some embodiments of the present disclosure. In some embodiments, the aerosol generating device 1 may be a square cylinder, to facilitate hand holding by a user. The aerosol generating device 1 is configured to bake and heat the aerosol generating substrate 2 inserted therein at the low temperature to release an aerosol extract from the aerosol generating substrate 2, and the atomization stability and the atomization taste are

good. In some embodiments, the aerosol generating substrate **2** may be a cylinder, and may be made of a string-shape or sheet-shape solid material that is made of leaves and/or stalks of a plant. In addition, a fragrance component may be further added to the solid material. It may be understood that the aerosol generating device **1** is not limited to being the square cylinder. In some other embodiments, the aerosol generating device **1** may alternatively be a cylinder, an elliptic cylinder, or in another shape.

[0046] In some embodiments, the aerosol generating device **1** may include a heating structure **10** and a housing **20** configured to carry the heating structure **10**. In some embodiments, the heating structure **10** may be a cylinder, and the aerosol generating substrate **2** may be detachably inserted into the heating structure **10**, so that the aerosol generating substrate **2** is heated and baked from the periphery. Alternatively, the aerosol generating substrate **2** is sleeved outside the heating structure **10**, so that the aerosol generating substrate **2** is heated and baked from the inside. In some embodiments, the aerosol generating device **1** may further include a power supply component disposed in the housing **20**. The heating structure **10** may be partially inserted into the aerosol generating substrate. Specifically, a part of the heating structure **10** may be inserted into a substrate section of the aerosol generating substrate **2**, and in a powered-on state, heat radiation is generated to heat the substrate section of the aerosol generating substrate **2**, so that the aerosol generating substrate **2** is atomized to generate an aerosol. In this embodiment, the heat radiation may be infrared radiation. The heating structure **10** has the advantages of being easy to assemble, simple structure, high atomization efficiency, high stability, and long service life. The power supply component is electrically connected to the heating structure **10**, to supply power to the heating structure **10**.

[0047] Refer to FIG. 2 and FIG. 3 together. In some embodiments, the heating structure **10** may include a tube body **11**, a heating element **12**, a first framework **13**, a second framework **14**, a third framework **15**, a fourth framework **16**, and a base **17**. The heating element **12** is disposed on the first framework **13**, the second framework **14**, the third framework **15**, and the fourth framework **16** in an M shape or an N shape. The heating element **12** is disposed in the tube body **11** together with the first framework **13**, the second framework **14**, the third framework **15**, and the fourth framework **16**. The tube body **11** covers at least a part of the heating element **12**, and allows an optical wave to pass through the aerosol generating substrate **2**. Specifically, in this embodiment, the tube body **11** may allow the infrared wave to pass through, so that the heating element **12** can radiate the infrared wave to heat the aerosol generating substrate **2**. Specifically, a spacing is defined between the heating element **12** and the tube body **11**. In the powered-on state, the heating element **12** rapidly heats up to 1000° C. to 1300° C. within 1 s to 3 s, the surface temperature of the tube body **11** may be controlled to be below 350° C., and the atomization temperature of the entire aerosol generating substrate **2** is controlled to 300° C. to 350° C., to implement precise atomization of the aerosol generating substrate **2** in a band of 2 μm to 5 μm. Specifically, the heating element **12** includes a heating part **121** that radiates an infrared wave in the powered-on state and two conductive parts **122**. The heating part **121** includes two heating sections. The two heating sections are respectively a first heating section **1211** and a second heating section **1212** connected to the first heating section **1211**. The first heating section **1211** includes a first end **1201** and a second end **1202** opposite to the first end **1201**. The second heating section **1212** includes a third end **1203** connected to the second end **1202** and a fourth end **1204** opposite to the third end **1203**. The two conductive parts **122** are separately connected to the fourth end **1204** of the second heating section **1212**. The first framework **13** and the second framework **14** are spaced apart and respectively disposed on the first end **1201** and the second end **1202**. The third framework **15** and the fourth framework **16** are spaced apart and respectively disposed on the third end **1203** and the fourth end **1204**. The first heating section **1211** is wound on the first framework **13** and the second framework **14** in an M shape or an N shape, and the second heating section **1212** is wound on the third framework **15** and the fourth framework **16** in an M shape or an N shape. The base **17** is disposed

at the lower open end of the tube body **11**, and the two conductive parts **122** are inserted into the base **17**, to fasten the heating element **12**. It may be understood that a quantity of the heating sections is not limited to two, and may be more than two, to match different aerosol generating substrates **2** and perform segmented heating on the aerosol generating substrates **2**. In some embodiments, the heating structure **10** may further include a support rod. The support rod is configured to connect the first framework **13**, the second framework **14**, the third framework **15**, and the fourth framework **16** to each other and is inserted into the base **17** to fasten the first framework **13**, the second framework **14**, the third framework **15**, and the fourth framework **16** to the inner surface of the tube body **11**. It may be understood that, in other embodiments, the first framework **13**, the second framework **14**, the third framework **15**, and the fourth framework **16** may alternatively be fastened to the inner surface of the tube body **11** in one of manners including direct bonding, clamping, or hot melting. The maximum working temperature of the heating element **12** ranges from 500° C. to 1300° C., which is much higher than 400° C. in the conventional technologies, so that problems such as burning easily and inconsistent taste in the high-temperature working environment are resolved, and the preheat time is greatly shortened.

[0048] In some embodiments, the tube body **11** may be a hollow tube. Specifically, the tube body **11** includes a tubular body **111** with a circular cross section and a pointed top structure **112** disposed at one end of the tubular body **111**. Certainly, it may be understood that, in some other embodiments, the cross section of the tubular body **111** is not limited to being circular. The tubular body **111** is of a hollow structure provided with an opening at one end. The tube body **11** may be mounted on the base **17**. Specifically, the tube body **11** may be partially inserted into the base **17**. The opening of the tube body **11** may be in the base **17**. The pointed top structure **112** is disposed at the end of the tubular body **111** away from the opening. Disposition of the pointed top structure **112** facilitates insertion of at least a part of the heating structure **10** into the aerosol generating substrate **2**. In this embodiment, a first accommodating cavity **113** is defined on the inner side of the tube body **11**. The first accommodating cavity **113** is a cylindrical cavity and may be unsealed. When the heating element **12** is mounted therein, the first accommodating cavity **113** does not need to be evacuated or filled with an inactive gas. It may be understood that, in some other embodiments, the heating element **12** may alternatively be disposed on and spaced apart from the outer periphery of the tube body **11**, and a second accommodating cavity for accommodating the aerosol generating substrate **2** may be defined on the inner side of the tube body **11**. In this embodiment, a spacing is defined between the inner wall of the tube body **11** and the heating element **12**, and the spacing may be filled with air or in a vacuum state. The spacing is defined, so that there is no direct contact between the tube body **11** and the heating element **12**.

[0049] In some embodiments, the first framework **13** may be a circular ring and may include a plurality of first wire slots **131** provided on the outer periphery edge of the first framework **13** at equal intervals. In some embodiments, the second framework **14** may be a circular ring and may include a plurality of second wire slots **141** that are provided on the outer periphery edge of the second framework **14** and that correspond to the plurality of first wire slots **131**. In some embodiments, the third framework **15** may be a circular ring and may include a plurality of third wire slots **151** provided on the outer periphery edge of the third framework **15**. In some embodiments, the fourth framework **16** may be a circular ring and may include a plurality of fourth wire slots **161** that are provided on the outer periphery edge of the fourth framework **16** and that correspond to the plurality of third wire slots **151**. The plurality of first wire slots **131** and the plurality of second wire slots **141** are used for the first heating section **1211** to be wound on, so that the first heating section **1211** is wound on the plurality of first wire slots **131** and the plurality of second wire slots **141** in an M shape or an N shape. The plurality of third wire slots **151** and the plurality of fourth wire slots **161** are used for the second heating section **1212** to be wound on, so that the second heating section **1212** is wound on the plurality of third wire slots **151** and the plurality of fourth wire slots **161** in an M shape or an N shape. In this way, the heating part **121** can

be disposed as a whole in a hollow manner, so that the heating element **12** has no central conduction risk, the heating element **12** is not caused to be partially unheated. In addition, there is no risk that the central heating part is covered by the outer ring heating part, so that the heat utilization of the heating element **12** is increased. It may be understood that the first heating section **1211** may alternatively be wound on the plurality of first wire slots **131** and the plurality of second wire slots **141** in another shape. Correspondingly, the second heating section **1212** may alternatively be wound on the plurality of third wire slots **151** and the plurality of fourth wire slots **161** in another shape. It may be understood that the first framework **13**, the second framework **14**, the third framework **15**, and the fourth framework **16** are not limited to being the circular ring and may alternatively be in another shape such as a square ring. The first framework **13**, the second framework **14**, the third framework **15**, and the fourth framework **16** are also not limited to being the ring and may alternatively be a cylinder.

[0050] In some embodiments, the first heating section **1211** and the second heating section **1212** are longitudinally disposed. The length of the first heating section **1211** is less than, equal to, or greater than the length of the second heating section **1212**, to match different aerosol generating substrates **2** and perform segmented heating.

[0051] In some embodiments, the first heating section **1211** and the second heating section **1212** include a plurality of M-shaped or N-shaped sections. The plurality of M-shaped or N-shaped sections are formed by winding or bending strip-shaped or linear heating wires. The distribution density of the plurality of M-shaped or N-shaped sections of the first heating section **1211** on the first framework **13** and the second framework **14** is lower than, equal to, or higher than the distribution density of the plurality of M-shaped or N-shaped sections of the second heating section **1212** on the third framework **15** and the fourth framework **16**. In this way, a plurality of external temperature fields are formed, to match a plurality of types of aerosol generating substrates **2** or to meet different power output requirements. The denser distribution of the M-shaped or N-shaped sections indicates the higher temperature of the temperature field.

[0052] In some embodiments, the spacing between the plurality of first wire slots **131** and the tube body **11** and the spacing between the plurality of second wire slots **141** and the tube body **11** include a plurality of configurations, and the spacing between the plurality of third wire slots **151** and the tube body **11** and the spacing between the plurality of fourth wire slots **161** and the tube body **11** include a plurality of configurations. The spacing between the tube body **11** and the heating element **12** may be adjusted by adjusting the spacing between the plurality of first wire slots **131** and the tube body **11** and the spacing between the plurality of second wire slots **141** and the tube body **11** and/or the spacing between the plurality of third wire slots **151** and the tube body **11** and the spacing between the plurality of fourth wire slots **161** and the tube body **11**, to avoid the excessively high local temperature of the heating element **12** caused by the heating element **12** being in contact with the tube body **11**. In addition, the temperature difference between the surface of the tube body **11** and the surface of the heating element **12** may be adjusted by adjusting the spacing, to avoid the scorching of the aerosol generating substrate **2**. In some embodiments, when the spacing between the plurality of first wire slots **131** and the tube body **11** is equal to the spacing between the plurality of second wire slots **141** and the tube body **11**, the spacing between the plurality of third wire slots **151** and the tube body **11** may be greater than, less than, or equal to the spacing between the plurality of fourth wire slots **161** and the tube body **11**. When the spacing between the plurality of first wire slots **131** and the tube body **11** is less than the spacing between the plurality of second wire slots **141** and the tube body **11**, the spacing between the plurality of third wire slots **151** and the tube body **11** may be greater than, less than, or equal to the spacing between the plurality of fourth wire slots **161** and the tube body **11**. When the spacing between the plurality of first wire slots **131** and the tube body **11** is greater than the spacing between the plurality of second wire slots **141** and the tube body **11**, the spacing between the plurality of third wire slots **151** and the tube body **11** may be greater than, less than, or equal to the spacing between

the plurality of fourth wire slots **161** and the tube body **11**.

[0053] Further, as shown in FIG. **4**, in some embodiments, the radial dimensions of the first framework **13** and the second framework **14** include a plurality of configurations, and the radial dimensions of the third framework **15** and the fourth framework **16** include a plurality of configurations. The radial dimensions of the first framework **13** and the second framework **14** and the radial dimensions of the third framework **15** and the fourth framework **16** may be set, so that a spacing between the upper end of the heating element **12** and the tube body **11** is different from a spacing between the lower end of the heating element **12** and the tube body **11**, to control the temperature field, so as to match different aerosol generating substrates **2** or devices. The denser wire winding indicates the higher temperature of the temperature field. In some embodiments, when the radial dimension of the first framework **13** is equal to the radial dimension of the second framework **14**, the radial dimension of the third framework **15** may be smaller than, equal to, or larger than the radial dimension of the fourth framework **16**. When the radial dimension of the first framework **13** is smaller than the radial dimension of the second framework **14**, the radial dimension of the third framework **15** may be smaller than, equal to, or larger than the radial dimension of the fourth framework **16**. When the radial dimension of the first framework **13** is greater than the radial dimension of the second framework **14**, the radial dimension of the third framework **15** may be smaller than, equal to, or larger than the radial dimension of the fourth framework **16**.

[0054] It should be noted that the heating element **12** includes a heating part and a conductive part. The heating part includes at least two heating sections arranged spaced apart. The two heating sections may be electrically connected by using a heating substrate, for example, a heating substrate of a straight section, or may be electrically connected by using another conductive material. In addition, the two heating sections may alternatively generate heat independently. To be specific, there is no electric connection between the two heating sections, and the two heating sections are controlled to generate heat separately or simultaneously based on a specific requirement. Similarly, the heating part may alternatively include three or more heating sections arranged at intervals. For a connection manner between the heating sections, refer to the foregoing implementations. Using a heating wire as an example, the at least two heating sections may be integrally formed by winding one heating wire, and the heating sections are electrically connected by a straight section having a particular length of the heating wire. In addition, each heating section may be formed by winding a separate heating wire. The heating sections may be electrically connected by using a resistive material, so that the heating sections are connected in series. Alternatively, the heating sections may not be electrically connected. Each heating section is separately connected to the conductive part, and the plurality of heating sections are connected in parallel. Through the foregoing disposing manner, segmented heating can be effectively implemented, and the temperature field is controlled to be distributed properly, so that the vaping taste is further improved.

[0055] In some embodiments, the heating element **12** may include an infrared radiation layer and a heating substrate that generates heat in the powered-on state. The heating substrate may generate heat in the powered-on state. The infrared radiation layer is disposed on the outer surface of the heating substrate and is used for radiating the infrared wave. In this embodiment, the heating substrate and the infrared radiation layer are distributed in a concentric circle on the cross section of the heating part.

[0056] In this embodiment, the heating substrate may be wound or bent to be a cylinder as a whole. Specifically, the heating substrate may be a heating wire. Certainly, it may be understood that, in some other embodiments, the heating substrate may not be limited to being the cylinder, and may alternatively be a sheet. In other words, the heating substrate may be a heating sheet. The heating substrate includes a metal substrate having the high-temperature oxidation resistance property. The metal substrate may be a metal wire. Specifically, the heating substrate may be a metal material

having good high-temperature oxidation resistance and high stability and not easily deformed, for example, a nickel-chromium alloy substrate (for example, a nickel-chromium alloy wire) or an iron-chromium aluminum alloy substrate (for example, an iron-chromium aluminum alloy wire). In this embodiment, the radial dimension of the heating substrate may range from 0.15 mm to 0.8 mm.

[0057] In this embodiment, the heating element **12** further includes an oxidation resistance layer, and the oxidation resistance layer is formed between the heating substrate and the infrared radiation layer. Specifically, the oxidation resistance layer may be an oxide film. The heating substrate is subject to high-temperature heat treatment, a dense oxide film is formed on the surface of the heating substrate, and the oxide film forms the oxidation resistance layer. Certainly, it may be understood that, in some other embodiments, the oxidation resistance layer is not limited to including the oxide film formed by itself. In some other embodiments, the oxidation resistance layer may be an oxidation resistance coating applied to the outer surface of the heating substrate. The oxidation resistance layer is formed, to ensure that the heating substrate is not oxidized or rarely oxidized when being heated in the air environment, thereby improving the stability of the heating substrate. Therefore, there is no need to evacuate the first accommodating cavity **113**, fill the first accommodating cavity **113** with an inactive gas or a reducing gas, and seal the opening, so that the entire process of assembling the heating structure **10** is simplified, and the manufacturing costs are reduced. In this embodiment, the thickness of the oxidation resistance layer may be selected from 1 μm to 150 μm . When the thickness of the oxidation resistance layer is less than 1 μm , the heating substrate is easily oxidized. When the thickness of the oxidation resistance layer is greater than 150 μm , the heat conduction between the heating substrate and the infrared radiation layer may be affected.

[0058] In this embodiment, the infrared radiation layer may be an infrared layer. The infrared layer may be formed on the side of the oxidation resistance layer away from the heating substrate by an infrared layer-forming substrate through high-temperature heat treatment. In this embodiment, the infrared layer-forming substrate may be silicon carbide, spinel, or a composite substrate thereof. Certainly, it may be understood that, in some other embodiments, the infrared radiation layer is not limited to being the infrared layer. In some other embodiments, the infrared radiation layer may be a composite infrared layer. In this embodiment, the infrared layer may be formed on the side of the oxidation resistance layer away from the heating substrate in manners including dip coating, spraying, and brush coating. The thickness of the infrared radiation layer may range from 10 μm to 300 μm . When the thickness of the infrared radiation layer ranges from 10 μm to 300 μm , the heat radiation effect thereof is good. Therefore, the atomization efficiency and the atomization taste of the aerosol generating substrate **2** are good. Certainly, it may be understood that, in some other embodiments, the thickness of the infrared radiation layer is not limited to ranging from 10 μm to 300 μm .

[0059] In some embodiments, the heating element **12** further includes a binding layer disposed between the oxidation resistance layer and the infrared radiation layer. The binding layer may be used for preventing local breakdown of the heating substrate, to further improve the binding force between the oxidation resistance layer and the infrared radiation layer. In some embodiments, a binder in the binding layer may be a glass powder. In other words, the binding layer may be a glass powder layer. In some implementations, the base **17** may be made of a material such as a ceramics insulation material and a PEEK high-temperature resistance insulation material. The base **17** may include two fixing through holes **171** provided on the base **17**. The two fixing through holes **171** allow insertion of the two conductive parts **122**.

[0060] FIG. 5 and FIG. 6 show a heating structure **10a** according to a second embodiment of the present disclosure. The heating structure **10a** mainly differs from the first embodiment in that in some embodiments, the heating structure **10a** may include a tube body **11a**, a heating element **12a**, a first framework **13a**, a second framework **14a**, a third framework **15a**, and a fourth framework

16a.

[0061] The heating element **12a** includes a heating part **121a** that generates heat radiation in a powered-on state and two conductive parts **122a**. The heating part **121a** includes two heating sections. The two heating sections are respectively a first heating section **1211a** and a second heating section **1212a** connected to the first heating section **1211a**. The first heating section **1211a** includes a first end **1201a** and a second end **1202a** opposite to the first end **1201a**. The second heating section **1212a** includes a third end **1203a** connected to the second end **1202a** and a fourth end **1204a** opposite to the third end **1203a**. The two conductive parts **122a** are separately connected to the fourth end **1204a**. The first framework **13a** and the second framework **14a** are spaced apart and respectively disposed on the first end **1201a** and the second end **1202a** in parallel. The third framework **15a** and the fourth framework **16a** are spaced apart and respectively disposed on the third end **1203a** and the fourth end **1204a** in parallel. The first heating section **1211a** is wound on the first framework **13a** and the second framework **14a** in an M shape or an N shape, and the second heating section **1212a** is wound on the third framework **15a** and the fourth framework **16a** in an M shape or an N shape. It may be understood that a quantity of the heating sections is not limited to two, and may be more than two, to match different aerosol generating substrates and perform segmented heating on the aerosol generating substrates. In some embodiments, the heating structure may further include a support rod. The support rod is configured to connect the first framework **13a**, the second framework **14a**, the third framework **15a**, and the fourth framework **16a** to each other. It may be understood that, in other embodiments, the first framework **13a**, the second framework **14a**, the third framework **15a**, and the fourth framework **16a** are fastened to the outer surface of the tube body **11a** in one of manners including direct bonding, clamping, and hot melting.

[0062] In some embodiments, the tube body **11a** may include a first sleeve tube **111a** and a second sleeve tube **112a** sleeved on the outer periphery of the first sleeve tube **111a**. The first sleeve tube **111a** is of a hollow structure with two open ends. The first sleeve tube **111a** may be a cylinder, and the inner diameter of the first sleeve tube **111a** may be slightly greater than the outer diameter of the aerosol generating substrate. A spacing is defined between the first sleeve tube **111a** and the second sleeve tube **112a**, and an accommodating cavity for accommodating the heating element **12a** is defined by the spacing. The axial length of the first sleeve tube **111a** may be greater than the axial length of the second sleeve tube **112a**. The second sleeve tube **112a** may be sleeved on the outer periphery of the first sleeve tube **111a**. The second sleeve tube **112a** may be a cylinder. The radial dimension of the second sleeve tube **112a** may be greater than the radial dimension of the first sleeve tube **111a**. In some embodiments, the heating element **12a** is wound on the outer periphery of the first sleeve tube **111a** and is spaced apart from the outer wall of the second sleeve tube **112a**, so that there may be a particular temperature difference between the inner wall of the accommodating cavity and the heating element **12a**, to achieve heat insulation. A heating cavity for heating the aerosol generating substrate is defined on the inner side of the first sleeve tube **111a**.

[0063] In some embodiments, the inner side of the second sleeve tube **112a** is provided with a reflective layer. The reflective layer is for reflecting heat from the heating element **12a** and radiating the heat to the aerosol generating substrate, to enhance the energy efficiency. It may be understood that the first sleeve tube **111a** and the second sleeve tube **112a** are not limited to being cylinder, and may alternatively be in another shape such as a square cylinder and an elliptical cylinder.

[0064] In some embodiments, the second sleeve tube **112a** may further include a fixing structure. The fixing structure is configured to fasten the heating element **12a**.

[0065] In some embodiments, the first framework **13a** may be a circular ring and may include a plurality of first wire slots **131a** provided on the outer periphery edge of the first framework **13a** at equal intervals. In some embodiments, the second framework **14a** may be a circular ring and may include a plurality of second wire slots **141a** that are provided on the outer periphery edge of the

second framework **14a** and that correspond to the plurality of first wire slots **131a**. In some embodiments, the third framework **15a** may be a circular ring and may include a plurality of third wire slots **151a** provided on the outer periphery edge of the third framework **15a** at equal intervals. In some embodiments, the fourth framework **16a** may be a circular ring and may include a plurality of fourth wire slots **161a** that are provided on the outer periphery edge of the fourth framework **16a** and that correspond to the plurality of third wire slots **151a**. The plurality of first wire slots **131a** and the second wire slots **141a** are used for the first heating section **1211a** to be wound on, so that the first heating section **1211a** is wound on the plurality of first wire slots **131a** and the plurality of second wire slots **141a** in an M shape or an N shape. The plurality of third wire slots **151a** and the plurality of fourth wire slots **161a** are used for the second heating section **1212** to be wound on, so that the second heating section **1212a** is wound on the plurality of third wire slots **151a** and the plurality of fourth wire slots **161a** in an M shape or an N shape. In this way, the heating part **121a** can be disposed in a hollow manner, so that the heating element **12a** has no central conduction risk, the heating element **12a** is not caused to be partially unheated. In addition, there is no risk that the central heating part is covered by the outer ring heating part, so that the heat utilization of the heating element **12a** is increased. It may be understood that the first heating section **1211a** may alternatively be wound on the plurality of first wire slots **131a** and the plurality of second wire slots **141a** in another shape. Correspondingly, the second heating section **1212a** may alternatively be wound on the plurality of third wire slots **151a** and the plurality of fourth wire slots **161a** in another shape. It may be understood that the first framework **13a**, the second framework **14a**, the third framework **15a**, and the fourth framework **16a** are not limited to being the circular ring and alternatively be in another shape such as a square ring. The first framework **13a**, the second framework **14a**, the third framework **15a**, and the fourth framework **16a** are also not limited to being the ring and may alternatively be a cylinder.

[0066] In some embodiments, the spacing between the plurality of first wire slots **131a** and the first sleeve tube **111a** and the spacing between the plurality of second wire slots **141a** and the first sleeve tube **111a** include a plurality of configurations, and the spacing between the plurality of third wire slots **151a** and the first sleeve tube **111a** and the spacing between the plurality of fourth wire slots **161a** and the first sleeve tube **111a** include a plurality of configurations. A spacing between the first sleeve tube **111a** and the heating element **12a** may be adjusted by adjusting the spacing between the plurality of first wire slots **131a** and the first sleeve tube **111a** and the spacing between the plurality of second wire slots **141a** and the first sleeve tube **111a** and/or the spacing between the plurality of third wire slots **151a** and the first sleeve tube **111a** and the spacing between the plurality of fourth wire slots **161a** and the first sleeve tube **111a**, to avoid the excessively high local temperature of the heating element **12a** caused by the heating element **12a** being in contact with the first sleeve tube **111a**. In addition, the temperature difference between the surface of the first sleeve tube **111a** and the surface of the heating element **12a** may be adjusted by adjusting the spacing, to avoid the scorching of the aerosol generating substrate. In some embodiments, when the spacing between the plurality of first wire slots **131a** and the first sleeve tube **111a** is equal to the spacing between the plurality of second wire slots **141a** and the first sleeve tube **111a**, the spacing between the plurality of third wire slots **151a** and the first sleeve tube **111a** may be greater than, less than, or equal to the spacing between the plurality of fourth wire slots **161a** and the first sleeve tube **111a**. When the spacing between the plurality of first wire slots **131a** and the first sleeve tube **111a** is less than the spacing between the plurality of second wire slots **141a** and the first sleeve tube **111a**, the spacing between the plurality of third wire slots **151a** and the first sleeve tube **111a** may be greater than, less than, or equal to the spacing between the plurality of fourth wire slots **161a** and the first sleeve tube **111a**. When the spacing between the plurality of first wire slots **131a** and the first sleeve tube **111a** is greater than the spacing between the plurality of second wire slots **141a** and the first sleeve tube **111a**, the spacing between the plurality of third wire slots **151a** and the first sleeve tube **111a** may be greater than, less than, or equal to the spacing between the plurality of fourth

wire slots **161a** and the first sleeve tube **111a**.

[0067] While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. It will be understood that changes and modifications may be made by those of ordinary skill within the scope of the following claims. In particular, the present invention covers further embodiments with any combination of features from different embodiments described above and below. Additionally, statements made herein characterizing the invention refer to an embodiment of the invention and not necessarily all embodiments.

[0068] The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

Claims

1. A heating structure, comprising: a heating element configured to radiate an infrared wave in a powered-on state; and a tube body disposed in conjunction with the heating element, the tube body being configured to allow the infrared wave to pass through, wherein the heating element is at least partially spaced apart from a tube wall of the tube body, wherein the heating element comprises a heating part and a conductive part, wherein the heating part comprises at least two heating sections arranged spaced apart, and wherein the heating sections are electrically connected to the conductive part.
2. The heating structure of claim 1, further comprising: a framework disposed on the tube body, wherein at least one of the heating sections is disposed on the framework.
3. The heating structure of claim 2, wherein the framework comprises a first framework, a second framework, a third framework, and a fourth framework, wherein the at least two heating sections comprise a first heating section and a second heating section connected to the first heating section, wherein the first framework and the second framework are spaced apart and separately disposed on the first heating section, and wherein the third framework and the fourth framework are spaced apart and separately disposed on the second heating section.
4. The heating structure of claim 2, wherein the heating part is longitudinally disposed, wherein the at least two heating sections comprise a first heating section and a second heating section connected to the first heating section, and wherein the first heating section and/or the second heating section are/is wound on the framework in an M shape or an N shape.
5. The heating structure of claim 4, wherein a length of the first heating section is less than, equal to, or greater than a length of the second heating section.
6. The heating structure of claim 4, wherein the first heating section and the second heating section comprise a plurality of M-shaped or N-shaped sections disposed at intervals.
7. The heating structure of claim 6, wherein the plurality of M-shaped or N-shaped sections are formed by winding or bending strip-shaped or linear heating wires, and wherein a distribution density of the plurality of M-shaped or N-shaped sections of the first heating section on the framework is lower than, equal to, or higher than a distribution density of the plurality of M-shaped

or N-shaped sections of the second heating section on the framework.

8. The heating structure of claim 3, wherein a radial dimension of the first framework is equal to a radial dimension of the second framework, and wherein a radial dimension of the third framework is smaller than, equal to, or larger than the radial dimension of the second framework.

9. The heating structure of claim 3, wherein a radial dimension of the first framework is smaller than a radial dimension of the second framework, and wherein a radial dimension of the third framework is smaller than, equal to, or larger than the radial dimension of the second framework.

10. The heating structure of claim 3, wherein a radial dimension of the first framework is larger than a radial dimension of the second framework, and wherein a radial dimension of the third framework is smaller than, equal to, or larger than the radial dimension of the second framework.

11. The heating structure of claim 3, wherein the first framework comprises a plurality of first wire slots provided on an outer periphery edge of the first framework, wherein the second framework comprises a plurality of second wire slots corresponding to the plurality of first wire slots, wherein the heating part is wound on the plurality of first wire slots and the plurality of second wire slots in an M shape or an N shape, wherein the third framework comprises a plurality of third wire slots provided on an outer periphery edge of the third framework, wherein the fourth framework comprises a plurality of fourth wire slots corresponding to the plurality of third wire slots, and wherein the heating part is wound on the plurality of third wire slots and the plurality of fourth wire slots in an M shape or an N shape.

12. The heating structure of claim 11, wherein, when a spacing between the plurality of first wire slots and the tube body is equal to a spacing between the plurality of second wire slots and the tube body, a spacing between the plurality of third wire slots and the tube body is greater than, less than, or equal to a spacing between the plurality of fourth wire slots and the tube body.

13. The heating structure of claim 11, wherein, when a spacing between the plurality of first wire slots and the tube body is less than a spacing between the plurality of second wire slots and the tube body, a spacing between the plurality of third wire slots and the tube body is greater than, less than, or equal to a spacing between the plurality of fourth wire slots and the tube body.

14. The heating structure of claim 13, wherein, when a spacing between the plurality of first wire slots and the tube body is greater than a spacing between the plurality of second wire slots and the tube body, a spacing between the plurality of third wire slots and the tube body is greater than, less than, or equal to a spacing between the plurality of fourth wire slots and the tube body.

15. The heating structure of claim 1, wherein the heating part comprises a heating substrate configured to generate heat in a powered-on state, and wherein an infrared radiation layer is disposed on an outer surface of the heating substrate and is configured to radiate the infrared wave.

16. The heating structure of claim 1, further comprising: a base, wherein the tube body is mounted on the base, and wherein the conductive part of the heating element passes through the base.

17. The heating structure of claim 1, wherein the heating element is disposed on an inner side of the tube body, and wherein a spacing is defined between the heating element and an inner wall of the tube body.

18. The heating structure of claim 11, wherein the tube body comprises a first sleeve tube and a second sleeve tube sleeved on an outer periphery of the first sleeve tube, wherein a spacing is defined between the first sleeve tube and the second sleeve tube, and an accommodating cavity configured to accommodate the heating element is defined by the spacing, wherein the heating element is disposed on and spaced apart from an outer periphery of the first sleeve tube, and wherein a heating cavity configured to heat an aerosol generating substrate is defined on an inner side of the first sleeve tube.

19. The heating structure of claim 18, wherein a reflective layer configured to reflect the infrared wave is disposed on an inner wall of the second sleeve tube.

20. The heating structure of claim 18, wherein when the spacing between the plurality of first wire slots and the first sleeve tube is equal to the spacing between the plurality of second wire slots and

the first sleeve tube, the spacing between the plurality of third wire slots and the first sleeve tube is greater than, less than, or equal to the spacing between the plurality of fourth wire slots and the first sleeve tube.

21. The heating structure of claim 18, wherein, when a spacing between the plurality of first wire slots and the first sleeve tube is less than a spacing between the plurality of second wire slots and the first sleeve tube, a spacing between the plurality of third wire slots and the first sleeve tube is greater than, less than, or equal to a spacing between the plurality of fourth wire slots and the first sleeve tube.

22. The heating structure of claim 18, wherein, when a spacing between the plurality of first wire slots and the first sleeve tube is greater than a spacing between the plurality of second wire slots and the first sleeve tube, a spacing between the plurality of third wire slots and the first sleeve tube is greater than, less than, or equal to a spacing between the plurality of fourth wire slots and the first sleeve tube.

23. An aerosol generating device, comprising: the heating structure of claim 1.
